PPS Perkins Needs Assessment: Aligning CTE with Priority Sectors

Regional Workforce Trends in Priority Sectors

Portland's economy is experiencing robust growth in four priority industries – **advanced manufacturing**, **construction**, **clean energy**, and **health care** – as identified by Worksystems. Key labor market trends include:

Advanced Manufacturing: The Portland metro is a national leader in high-tech manufacturing (e.g. semiconductors) and related industries. The sector employs over 70,000 workers regionally, with an average wage above $130k (worksystems.org). Although five-year job growth has been steady at 5% (worksystems.org), output has surged (gross regional product up 85% since 2019), reflecting increasing productivity and investment, worksystems.org. Emerging skill needs center on semiconductor manufacturing and advanced machining – fueled by soaring demand and federal CHIPS Act investments, worksystems.org– as well as automation and robotics to drive innovation. Employers report strong demand for technicians (e.g. semiconductor processing techs, machinists, welders) with wages in the ~$27–32/hr range (worksystems.org), indicating a need for a talent pipeline in these trades.

Construction: The construction industry in the Portland-Vancouver metro area is booming, adding over 28,000 jobs since 2015 (about 40% growth) (worksystems.org). 2023 it accounted for 51,600 jobs, with an average annual wage of over $105k (worksystems.org). Construction employment will continue rising (another ~11,500 high-paying jobs by 2032) amid major infrastructure and development projects (worksystems.org). Skilled trades are especially in demand – carpenters (6,900+ jobs), laborers (5,500+ jobs), electricians (~4,600 jobs), and plumbers (~2,600 jobs) are among the top occupations, often earning $25–47 per hour (worksystems.org). An aging workforce (many nearing retirement) and ongoing public investments have created an urgent need for new trade entrants (worksystems.org). Industry stakeholders are emphasizing apprenticeships and training programs to meet the labor shortfall in construction.

Clean Energy: A regional transition to clean, renewable energy is underway, driving cross-sector workforce needs. Broadly defined, the clean energy sector (including renewable energy generation, energy efficiency, and related green construction/manufacturing roles) employs about 139,500 workers in the metro area (worksystems.org) – a high concentration (LQ 1.8) relative to the nation. With an average wage of $114,725, these roles provide an opportunity for economic mobility for many students. Five-year job growth is ~8% (worksystems.org) and expected to accelerate as Oregon invests in sustainable infrastructure (solar, wind, energy-efficient buildings, electric transportation). Industry demands span

multiple disciplines: construction trades (e.g. electricians, HVAC installers, solar installers) to build and retrofit green buildings (worksystems.org), manufacturers of clean technology (batteries, electric vehicles, etc.), and technicians to operate and maintain renewable energy systems. Key employers range from utilities (PGE, Pacific Power) to clean-tech firms (e.g. Vestas, ESS Inc) (worksystems.org). This sector's growth calls for workers with traditional skills (e.g. electrical, carpentry, with a green focus) and new skills in emerging technologies like solar PV, energy storage, and electric vehicle infrastructure.

Health Care: The health care and social assistance sector is expanding rapidly in Portland, driven by an aging population and rising demand for health services (worksystems.org). Regional health care employment tops 109,700 jobs with 9% growth in the past five years (worksystems.org), and an average wage around $81k (worksystems.org). Major hospital systems (Legacy, Providence, Kaiser, OHSU) are among the largest employers (worksystems.org). Emerging skill needs include not only traditional roles like nurses and medical assistants, but also behavioral health professionals and medical office/support staff. Behavioral health is a critical growth area – the region faces a shortage of qualified mental health and substance abuse support workers (worksystems.org). Many health careers offer entry-level opportunities without a four-year degree; nearly 60% of health jobs paying $22+/hr do not require a bachelor's (worksystems.org) Growing occupations include medical assistants, health information technicians, licensed practical nurses, and peer support specialists (worksystems.org). High retirement rates (and burnout from the pandemic) further intensify the need to train new healthcare workers across clinical and support roles (worksystems.org)

Current CTE Offerings and Gaps Analysis

Portland Public Schools offers a broad array of Career & Technical Education programs, with 78 state-approved CTE programs of study across 11 high school campuses. These programs span six career clusters – including Health Sciences and Industrial & Engineering Systems, which correspond to the priority sectors – and fields like Business, Arts/Communication, Agriculture/Natural Resources, and Human Services. This breadth provides many students with exposure to technical skills, but alignment with regional workforce needs in the four high-growth sectors is uneven.

Alignment with Priority Sectors: Some PPS programs are well-aligned with local industry demand. For example, the district's Health Sciences pathways (e.g. medical professions or bio-science programs at certain high schools) feed into the healthcare talent pipeline by teaching anatomy, patient care, and even offering certifications like CNA or First Aid. Similarly, several Industrial & Engineering Systems programs cover skills relevant to manufacturing and construction – such as welding, engineering design, automotive technology, and construction trades – which map to advanced manufacturing and construction jobs. Flagship career-tech schools like Benson Polytechnic High School historically offer manufacturing technology, electrical, architectural design, and automotive programs, producing graduates who often enter apprenticeships or technical college in related fields. These existing programs provide a foundation to build, indicating that PPS has some infrastructure for the targeted industries.

However, significant gaps exist between current offerings and emerging workforce needs in the priority sectors:

Advanced Manufacturing: Despite the region's leadership in high-tech manufacturing, PPS has limited CTE programs explicitly focused on modern manufacturing technology (e.g. semiconductor fabrication, robotics, precision machining). Traditional manufacturing classes (welding, wood/metal shop or basic engineering) exist, but advanced manufacturing pathways that incorporate electronics, automation, and clean-room processes are sparse. There is an opportunity to introduce or expand mechatronics, electronics fabrication, or industrial automation programs to meet the growing demand for semiconductors and advanced manufacturing workforces (worksystems.org). Without such programs, employers in the booming high-tech manufacturing arena may face a shortage of entry-level technicians familiar with industry-grade equipment and processes.

Construction Trades: PPS offers construction technology or woodworking programs at only a few high schools, meaning many students lack access to this career pathway. Given the urgent demand for skilled construction labor region-wide (worksystems.org), the current scale of construction-related CTE programs may be insufficient. For instance, if only one or two campuses have building trades programs, the output of graduates ready for apprenticeships is relatively low. Gaps include specialty trades exposure (plumbing, HVAC, electrical) – students may not get hands-on experience in these areas unless they attend the one school with a construction program. To meet employer demand, new or expanded programs (or magnet initiatives) in the building trades could be warranted, so that students across the district can prepare for high-wage construction careers.

Clean Energy: There is currently *no dedicated "clean energy" CTE program* in PPS, even though sustainability and energy technology are fast-growing fields. Clean energy elements are touched on indirectly in other programs (for example, an environmental science course or a unit on solar power in a construction class). Still, a coherent program of study focused on renewable energy, energy efficiency, or green technology is lacking. This represents a gap in preparing students for the burgeoning green economy. A Renewable Energy Technology program – covering solar panel installation, wind/energy generation, battery storage, and green building design – could fill this void and align directly with Portland's climate and infrastructure initiatives. Likewise, automotive technology programs could expand to include electric vehicle (EV) maintenance and charging infrastructure, bridging into clean transportation careers. Without explicitly addressing clean energy skills, PPS is missing an opportunity to train students for a sector employing over 139,000 locally (worksystems.org) and growing in importance.

Healthcare & Social Assistance: PPS's health science programs are a strength, but they could be broadened to address emerging needs like behavioral health and healthcare administration. For example, existing programs often focus on general medical or biomedical themes; there is room to add mental and behavioral health support pathways, given the acute regional demand for those roles (worksystems.org). Additionally, internships or clinical experiences in hospitals and clinics (e.g. through partnerships with major healthcare providers) are limited by capacity – more opportunities are needed so interested students districtwide can

get real-world healthcare experience. Another gap is in continuity to postsecondary: strengthening connections to nursing programs or allied health training at local colleges would help students seamlessly transition into the healthcare workforce pipeline. Ensuring every high school has some offering in the health sciences (or access via cross-enrollment) would improve equity of access to this high-demand field.

In summary, while PPS's 79 CTE programs cover many career fields, they do not fully reflect the current labor market priorities. High-growth sectors like advanced manufacturing, clean energy, and construction are under-represented in the CTE portfolio, signaling a need for new programs or expansion. Strengthening these areas (without neglecting existing popular programs) will better align student skills with employer demand and address gaps where local industries are clamoring for talent.

Investment Strategies & Partnership Integration for Industry Alignment

PPS can leverage its current Perkins-funded initiatives and partnership models to ensure CTE program growth aligns with regional industry needs. Current investment areas under Perkins V – such as Program Advisory Councils, Work-Based Learning, curriculum development, and industry-recognized credentials – provide a foundation for strengthening the connection between education and the workforce. The district's strategy is to go beyond standalone classes by involving industry and community partners in CTE. Key approaches include:

Program Advisory Councils (PACs): Every CTE program convenes industry advisors who guide curriculum and program improvements. PPS can maximize PAC impact by recruiting representatives from the priority sectors (advanced manufacturing, construction, clean energy, healthcare) to these councils. Regular consultation with employers ensures that programs stay up-to-date with evolving industry needs and technologies. For example, industry advisors in manufacturing can suggest emerging skills (like CNC programming or quality control standards) to incorporate into coursework. In healthcare, hospital partners on advisory boards can help align high school training with entry-level job requirements. Strengthening PACs and giving them a strategic role in program design will keep career pathways relevant and demand-driven.

Work-Based Learning (WBL): PPS already invests Perkins funds to expand work-based learning opportunities such as internships, job shadows, and project-based learning with industry. These efforts directly support alignment by giving students hands-on experience in high-demand fields. For instance, internships with local manufacturing firms or hospitals allow students to apply skills in real workplaces and understand industry expectations. The district's Partnership Engagement Menu outlines numerous ways businesses can engage – from offering industry projects for students to providing internships and mentorship. By leveraging this menu and coordinating through College & Career Centers, PPS can scale up WBL in the four target sectors. Emphasizing quality WBL (with training plans and reflection) helps students build the "soft skills" and technical competencies employers seek, making graduates more work-ready.

Curriculum Development & Industry Credentials: Ongoing curriculum modernization is funded in part by Perkins to ensure programs integrate new technologies and standards. PPS can channel these funds to update curricula aligned with industry 4.0 practices (e.g. adding modules on renewable energy in science or CAD/CNC in manufacturing classes). Collaboration with local community colleges (many of which receive Perkins funds too) can create alignment between high school programs and college technical courses, ensuring a smoother student transition In addition, supporting students in earning industry-recognized credentials is a priority – for example, OSHA-10 safety certification in construction, Certified Nursing Assistant (CNA) or First Responder certifications in health, or AutoCAD and SolidWorks certifications in engineering. These credentials give students tangible proof of skills employers recognize, thereby tightening the linkage between CTE outcomes and workforce entry. PPS's investments in credential exam fees, teacher training, and equipment all contribute to programs that meet industry standards.

Cross-Functional Partnership Teams: Beyond traditional advisory roles, PPS has pioneered a *cross-functional team approach* to CTE partnerships. This model brings educators, industry professionals, workforce agencies, and community organizations together as a collaborative team to solve workforce challenges. Instead of isolated, one-off partnerships, cross-functional teams work on coordinated strategies – for example, a healthcare team might include teachers, hospital HR staff, a community college nursing instructor, and a workforce board representative working collectively to expand a pathway for future medical assistants. By working in tandem, these stakeholders can align curriculum, work-based experiences, and hiring practices to ensure students graduate with in-demand skills. The power of this approach is that it moves beyond transactional engagements (like a company donating equipment) toward deeper co-development of programs. Cross-functional teams focus on shared goals such as diversifying the talent pipeline or updating training for emerging technologies, and they iterate solutions together. PPS's use of this model (supported by its CTE team)will enhance employer engagement and keep programs responsive to economic shifts. For example, a cross-sector clean energy team could help design a new renewable energy CTE program, with input on everything from curriculum to internship placements. This integrated partnership strategy ensures sustainable alignment, as solutions are developed collaboratively and are more likely to endure changes in staffing or funding.

Pathway Ecosystem Model: PPS also adopts a Pathway Ecosystem framework, which envisions every CTE pathway supported by a network of partners (industry, post-secondary, nonprofits, government, cultural institutions) working in concert. This systemic approach, funded partly through Perkins innovation grants and local initiatives, ensures students get layered support and real-world exposure throughout their journey. For instance, a health science pathway in an ecosystem might include a university dual-credit agreement, a hospital internship program, a nonprofit providing career mentors, and a government workforce agency supplying labor market data. Such an ecosystem makes the career pathway more sustainable, equitable, and adaptable. It helps institutionalize partnerships so that even if a champion leaves, the path continues with multiple stakeholders invested. Students benefit through consistent industry interaction (not just a single job shadow but ongoing engagement) and smoother transitions to college or apprenticeships. By integrating this model, PPS can leverage

cross-sector resources – for example, using Worksystems labor market reports to inform programs, or tapping state apprenticeship agencies to connect students with apprenticeships – thereby fully aligning CTE programs with regional workforce development efforts.

In summary, PPS's current Perkins investments in advisory councils, work-based learning, curriculum, and credentials can be strategically **deployed to bolster industry alignment**, especially when combined with innovative partnership structures. Through substantial advisory input, experiential learning, up-to-date curricula, and collaborative cross-functional teams, the district can ensure its CTE programs directly support the talent needs of Portland's growth sectors while providing students with relevant, work-ready skills.

Actionable Recommendations for Alignment and Growth

To maximize CTE alignment with regional economic needs, PPS should undertake targeted actions in program development, partnerships, and resource allocation. Below are strategic recommendations to guide program expansion, funding, and partnerships – all with an emphasis on enhancing student access, equity, and long-term pathway sustainability in advanced manufacturing, construction, clean energy, and healthcare:

Expand High-Demand CTE Pathways: Allocate funding to develop new programs and enhance capacity in the priority sectors. For example, establish a Renewable Energy Technology program (focused on solar, wind, and energy efficiency) to directly address the clean energy workforce gap, and consider a specialized Advanced Manufacturing or mechatronics program in partnership with a community college or industry consortium (leveraging the semiconductor initiative momentum (worksystems.org)). Similarly, construction trades programs could be expanded to additional high schools or created regional trades skill centers so more students can access carpentry, electrical, and plumbing training. Add a Behavioral Health/Patient Care pathway to prepare students for roles like medical assistant, phlebotomist, or peer recovery mentor, tapping into local hospital partnerships. Direct a significant portion of Perkins V and local CTE funds toward these new or expanded programs (for equipment, teacher training, and facility needs), as these fields align with Portland's highest-growth jobs. By increasing CTE slots in these areas, PPS will produce more graduates with skills matching critical job openings, supporting student success and regional workforce needs.

Strengthen Industry & Education Partnerships: Deepen collaboration with employers, labor unions, post-secondary institutions, and workforce boards to ensure programs remain tightly aligned with industry trends. Form sector-specific advisory committees or industry roundtables for each priority sector (if not already in place) – e.g. a Healthcare Advisory Panel with local hospital and clinic representatives, a Manufacturing Advisory with tech manufacturers like Intel or local fabrication shops, etc. – to provide ongoing feedback and pledges of support (guest instructors, site visits, donated equipment). Implement the cross-functional team approach by convening working groups for each sector that include CTE teachers, industry experts, Worksystems or workforce agency staff, and community partners to

co-design solutions (such as updating curriculum or coordinating internships). Seek out partnership agreements with organizations already active in these workforce areas: for instance, partner with union apprenticeship programs in construction (electricians, carpenters unions) to create a direct bridge from high school to apprenticeship, or with primary healthcare providers to guarantee clinical internship slots for students. Leverage Worksystems' support (as the regional workforce board) to connect with their initiatives – for example, collaborate on utilizing the Portland Clean Energy Fund grants for youth training in clean energy jobs (greaterportlandinc.com). PPS can integrate real-world industry expertise into CTE by formalizing and expanding these partnerships. This will yield benefits like curriculum co-development, teacher externships in industry, and a pipeline for students to jobs or further training. Employer engagement should be a key performance metric – e.g., each program of study in a priority sector should have at least one active industry partner per student, and each partner should commit to specific activities (advisory participation, hosting interns, etc.). These partnerships will keep CTE programs agile and demand-driven.

Enhance Work-Based Learning & Credentials: Increase work-based learning opportunities and embed industry credentials in all priority-sector programs to improve student readiness for high-wage careers. Work with industry and community partners to increase the number of internships, apprenticeships, and job shadow placements available to CTE students. For example, expand the SummerWorks or similar internship programs in coordination with Worksystems so that every student in an advanced manufacturing or construction pathway gets a paid summer internship or apprenticeship exposure before graduation (worksystems.org). Integrate pre-apprenticeship models into high school programs – for instance, align construction programs with BOLI-certified pre-apprenticeship curricula so students can directly enter apprenticeships with advanced standing. Ensure students have access to earn relevant certifications: OSHA 10 and Forklift Operator (for construction/manufacturing safety), CAD/CAM software certifications for engineering, NABCEP entry-level solar installer certification for a future clean energy program, and Basic Life Support or Certified Medical Assistant credentials for health students. Perkins funds can subsidize certification exam fees and support teacher professional development to implement these credentials.

Additionally, coal-credit agreements should be considered at th local colleges (like Portland Community College) so that students in these CTE programs can earn college credit or even complete the first term of a technical certificate while in high school. By graduation, students could have both a high school diploma and industry-recognized credentials or credits, dramatically improving their employability and postsecondary transition. Making work-based learning a graduation expectation for CTE concentrators (with adequate support to find placements) and tracking credential attainment will help institutionalize this focus. These efforts will produce a workforce-ready graduate with practical experience and verifiable skills aligning with industry requirements.

Promote Access, Equity & Pathway Sustainability: Ensure that the expansion of CTE opportunities in these sectors is equitable and sustainable for the long term. Intentional outreach and support are needed to include historically underrepresented students (girls, students of color, low-income students, students with disabilities) in high-demand programs

like manufacturing, construction, and STEM fields. Partner with community organizations with equity expertise to improve recruitment and retention – for example, collaborate with Oregon Tradeswomen, Constructing Hope, and Portland YouthBuilders to engage female and BIPOC youth in construction trades (worksystems.org) These organizations can provide role models, workshops, and pre-entry training that build confidence and interest among underrepresented students. Similarly, culturally responsive strategies and wrap-around supports (career coaches, mentorship programs, transportation assistance, childcare for teen parents, etc.) should be utilized to minimize barriers to participation (greaterportlandinc.com). On the sustainability front, advocate for stable funding streams: prioritize Perkins and state CTE funds toward these high-growth sector programs in future budgets, and pursue supplementary grants (such as Future Ready Oregon or federal innovation grants) to invest in equipment and teacher training. Develop a teacher talent pipeline for these programs by offering current industry professionals a pathway into teaching (e.g., through restricted CTE licensure programs) (pps.net) and supporting existing teachers with externships to keep skills current. Finally, embed the partnership ecosystem approach into policy so that each program has multiple partners co-invested in its success – this distributed support will make programs resilient to changes. By focusing on equity and sustainable practices, PPS can ensure that *all* students have access to these high-opportunity pathways and that the programs will thrive for years to come, continually adapting to the region's economic evolution.

These recommendations provide a roadmap for aligning Portland Public Schools' CTE offerings with the region's priority workforce sectors. BPPS can create a future-ready CTE ecosystem by strategically expanding programs, tightening partnership linkages, enriching work-based learning, and prioritizing equity and sustainability, Such an ecosystem will produce well-prepared graduates for careers in advanced manufacturing, construction, clean energy, and healthcare – thereby meeting employer demands and fostering economic growth, empowering students with the skills and opportunities to succeed in high-wage, high-demand industries.